

BHAVESH ANANDPARA

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EXPERIENCE

Associate Software Engineer at Bentley Systems

06/2024 - Current

- Built an agentic automation layer using **MCP, Code Mode, and GitHub Copilot** across **15+ drawing-production tools**, enabling users to dynamically compose and execute complex, multi-step engineering workflows through natural-language instructions.
- Adopted an **AI-first software development workflow** using **Claude Code, context engineering, reusable Skills, and task-specific agents** for codebase exploration, implementation, pre-PR reviews, architectural validation, and automated test generation, accelerating development while preserving human-in-the-loop quality control.
- Built **3 GitHub Actions workflows** automating security vulnerability remediation, Doppelganger dependency validation, and release-branch synchronization, shifting critical checks to pre-merge automation and preventing dependency issues that previously required up to **half a day** of debugging and remediation.
- Modernized **90+ static WPF dialogs** using **C#/.NET, WPF, and Prism**, implementing reusable horizontal, vertical, and bidirectional resizing with screen-aware constraints to improve usability across varying display resolutions.
- Developed and validated Python wrappers for **80+ C#/gRPC APIs**, exposing core application capabilities to Python and enabling script-based automation of engineering workflows with **100% of implemented wrappers tested**.
- Automated parallel-road mapping across multiple ETL data sources using a **Python/QGIS plugin**, reducing manual database-update effort by approximately **30%** and preventing an estimated **90%+ of cross-source data inconsistencies** caused by manual updates.
- Built and evaluated a **VGG16 transfer-learning model** for bridge detection from satellite imagery, achieving **89.7% validation accuracy** and approximately **0.90 precision, recall, and F1-score** across the evaluation set.
- Developed the end-to-end **Image View module using React and TypeScript**, applying **memoization, custom hooks, and reusable component patterns** to optimize rendering, enabling users to visualize IoT sensor data on 2D imagery without requiring a 3D model while achieving **76% automated test coverage**.

DevOps Intern at Coditas Solutions LLP

01/2024 - 06/2024

- Deployed cloud infrastructure using **AWS EC2, S3, VPC, ALB, Auto Scaling, CloudWatch, and SNS**, implementing load balancing, horizontal scaling, monitoring, and event-driven alerting for scalable application environments.
- Built **Jenkins CI/CD pipelines** and containerized applications using **Docker**, automating build and deployment workflows and reducing manual release effort.

SKILLS

Languages	C#, TypeScript, JavaScript (ES6+), Python, Java, HTML, CSS
Frontend	React.js, Redux, Three.js, Tailwind CSS
Backend	.NET, Node.js, Express.js, Flask, REST APIs
Databases	MySQL, MongoDB, Firebase, Neo4j, Qdrant
Cloud & DevOps	AWS, Docker, Jenkins, GitHub Actions, CI/CD
Testing	Playwright, Jest, Postman
AI & Developer Productivity	MCP, Agentic Workflows, Multi-Agent Systems, Context Engineering, GitHub Copilot, Prompt Chaining, Guardrails, Claude Code, Claude Skills, Task-Specific Agents

PROJECTS

CodeAtlas – AI-Powered Software Architecture Intelligence Platform

Personal Project

- Built a full-stack platform using **React, TypeScript, Node.js, and Express** that analyzes public GitHub repositories using **AST-based static code analysis** to extract code structure, dependencies, routes, and relationships.
- Architected a modular backend following **Clean Architecture, SOLID, and dependency inversion**, separating repository analysis, AST parsing, dependency graphs, semantic search, documentation generation, and AI reasoning into independent services.
- Built a hybrid retrieval architecture using **Neo4j knowledge graphs, Qdrant vector search, and RAG**, combining structural code relationships with semantic retrieval to power repository-aware search, architecture documentation, Mermaid diagrams, and dependency impact analysis.
- Designed a **provider abstraction and Strategy-based integration** supporting **Ollama, OpenAI, Claude, and Gemini**, decoupling model providers from core application logic and allowing models to be switched without modifying downstream services.

EDUCATION

Bachelor of Technology, in Computer Engineering

Bajaj Institute of Technology, Wardha.

2024

8.57 CGPA